Lakshya JEE 2.0 2025

Maths

Matrices

DPP: 1

Q1 If $A=\begin{bmatrix}5 & -2 & 6 & 1\\ 7 & 0 & 8 & -3\\ \sqrt{2} & \frac{3}{5} & 4 & 3\end{bmatrix}$ then write:

(i) The number of rows in A

(ii) The number of column in A

(iii) The order of the matrix A

(iv) The number of all entries in A

(v) The elements $a_{23},\ a_{31},\ a_{14},\ a_{33}$, a_{22} of A

Q2 Write the order of each of the following matrices

(i) $A=\begin{bmatrix}3 & 5 & 4 & -2\\ 0 & \sqrt{3} & -1 & \frac{4}{9}\end{bmatrix}$

(ii) $B=\begin{bmatrix}6 & -5\\ \frac{1}{2} & \frac{3}{4}\\ -2 & -1\end{bmatrix}$

(iii) $\mathrm{C}=\begin{bmatrix}7 & -\sqrt{2} & 5 & 0\end{bmatrix}$

(iv) $\mathrm{D}=[8,-3]$

(v) $E=\begin{bmatrix}-2\\ 3\\ 0\end{bmatrix}$

(vi) $\mathrm{F}=[6]$

Q3 Construct a 3×2 matrix whose elements are given by $a_{ij}=(2i-j)$.

Q4 Construct a 4×3 matrix whose elements are given by $a_{ij}=\frac{i}{j}$.

Q5 The matrix $\begin{bmatrix}2 & 0 & 0\\ 0 & 3 & 0\\ 0 & 0 & 3\end{bmatrix}$ is

(A) Column matrix

(B) Diagonal matrix

(C) Identity matrix

(D) Null matrix

Q6 The matrix $\begin{bmatrix}-1 & 0 & 0\\ 2 & -1 & 0\\ 3 & -7 & -1\end{bmatrix}$ is

(A) Upper triangular matrix

(B) Diagonal matrix

(C) Scalar matrix

(D) Lower triangular matrix

Q7 If a matrix has 18 elements, what are the possible orders it can have?

Q8 If $A=\begin{bmatrix}1 & -5 & 7\\ 0 & 7 & 9\\ 11 & 8 & 9\end{bmatrix}$, then trace of matrix A is:

(A) 17 (B) 25

(C) 3 (D) 12

Q9 If $x\begin{bmatrix}-3\\ 4\end{bmatrix}+y\begin{bmatrix}4\\ 3\end{bmatrix}=\begin{bmatrix}10\\ -5\end{bmatrix}$, then:

(A) $x=-2, y=1$

(B) $x=-9, y=10$

(C) $x=22, y=1$

(D) $x=2, y=-1$

Q10 If $1, \omega$ and ω^2 are the cube roots of unity and $\begin{bmatrix}1+\omega & 2\omega\\ -2\omega & -b\end{bmatrix}+\begin{bmatrix}a & -\omega\\ 3\omega & 2\end{bmatrix}=\begin{bmatrix}0 & \omega\\ \omega & 1\end{bmatrix}$ then a^2+b^2 is equal to:

(A) $1+\omega^2$

(B) ω^2-1

(C) $1+\omega$

(D) $(1+\omega)^2$

Q11 If $m[-3 \quad 4]+n[4 \quad -3]=[10 \quad -11]$, then $3m+7n$ is equal to:

(A) 3 (B) 5

(C) 10 (D) 1

Q12 If $A=\begin{bmatrix}1 & 2 & x\\ 3 & -1 & 2\end{bmatrix}$ and $B=\begin{bmatrix}y\\ x\\ 1\end{bmatrix}$ be such that $AB=\begin{bmatrix}6\\ 8\end{bmatrix}$, then

(A) $y=2x$ (B) $y=-2x$

(C) $y=x$ (D) $y=-x$

Q13 If $U=[2 \quad -3 \quad 4], X=[0 \quad 2 \quad 3], V$ and $=\begin{bmatrix}3\\ 2\\ 1\end{bmatrix}$

$Y=\begin{bmatrix}2\\ 2\\ 4\end{bmatrix}$, then $UV+XY$ is equal to:

(A) $[20]$

(B) 20

(C) $[-20]$

(D) -20

Q14 If $A=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ a & b & -1\end{bmatrix}$ then A^2 is equal to:

(A) 0

(B) $-A$

(C) I

(D) $2A$

Q15 If $A=\begin{bmatrix}2 & -1\\ 1 & 2\end{bmatrix}$, then $A^2+2A-3I$ is equal:

(A) $\begin{bmatrix}4 & -6\\ 6 & 4\end{bmatrix}$

(B) 0

(C) $\begin{bmatrix}-6 & 2\\ -2 & 6\end{bmatrix}$

(D) $5I$

Q16 If $A=\begin{bmatrix}x & 1\\ 1 & 0\end{bmatrix}$, and A^2 is the identity matrix, then x is equal to

(A) -1 (B) 0

(C) 1 (D) 2

Q17 If $[1 \quad x \quad 1]\begin{bmatrix}1 & 2 & 3\\ 0 & 5 & 1\\ 0 & 3 & 2\end{bmatrix}\begin{bmatrix}x\\ 1\\ -2\end{bmatrix}=0$, then the value of x is:

(A) 0

(B) $2/3$

(C) $\frac{5}{4}$

(D) $-4/5$

Q18 If $A=\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix}$ and I is the unit matrix of order 2, then A^2 is equal to

(A) $4A-3I$

(B) $3A-4I$

(C) $A-I$

(D) $A+I$

Answer Key

Q1 (i) 3

(ii) 4

(iii) 3×4

(iv) 12

(v) $a_{23} = 8,\ a_{31} = \sqrt{2}\ ,\ a_{14} = 1,\ a_{33} = 4,$ $a_{22} = 0$

Q2 (i) 2×4

(ii) 3×2

(iii) 1×4

(iv) 1×2

(v) 3×1

(vi) 1×1

Q3 $\begin{bmatrix} 1 & 0 \\ 3 & 2 \\ 5 & 4 \end{bmatrix}$

Q4 $\begin{bmatrix} 1 & 1/2 & 1/3 \\ 2 & 1 & 2/3 \\ 3 & 3/2 & 1 \\ 4 & 2 & 4/3 \end{bmatrix}$

Q5 (B)

Q6 (D)

Q7 So the order of the matrix can be $1 \times 18,\ 2 \times 9,\ 3 \times 6,\ 6 \times 3,\ 9 \times 2$ or 18×1

Q8 (A)

Q9 (A)

Q10 (C)

Q11 (D)

Q12 (A)

Q13 (A)

Q14 (C)

Q15 (A)

Q16 (B)

Q17 (C)

Q18 (A)

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